KYUNGDONG UNIVERSITY

경동대학교

WEEKLY PROGRESS REPORT

WEEK 5

Model Architecture & Initial Training

Presented By

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Course: Computer Vision | Professor Sarwar | KDU

5.1 Work Completed This Week

Week 5 covered the model build and the first successful training run. After multiple failed attempts with EfficientNetV2M due to model size and convergence issues, the team switched to EfficientNetV2S and achieved 55% accuracy in epoch 1, confirming the pipeline fix from Week 4 was correct. Phase 1 training on clean images completed with 78.03% validation accuracy.

Model Architecture

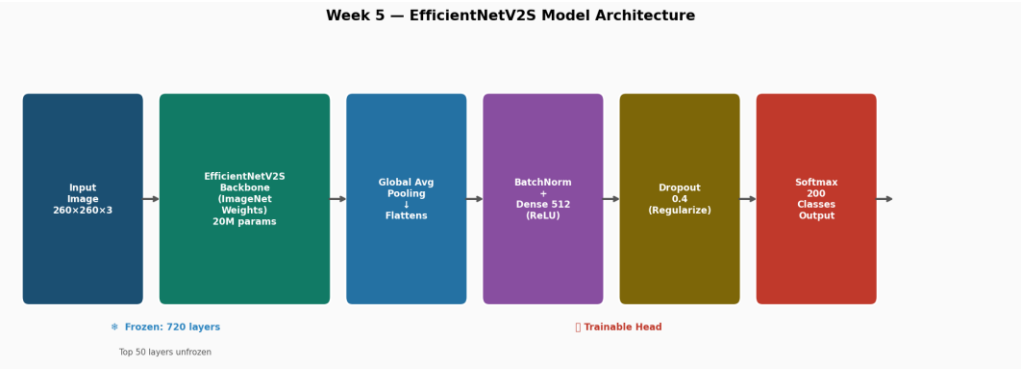

Figure 3: EfficientNetV2S architecture — backbone frozen except top 50 layers, custom classification head for 200 species

Problems Encountered & Solutions

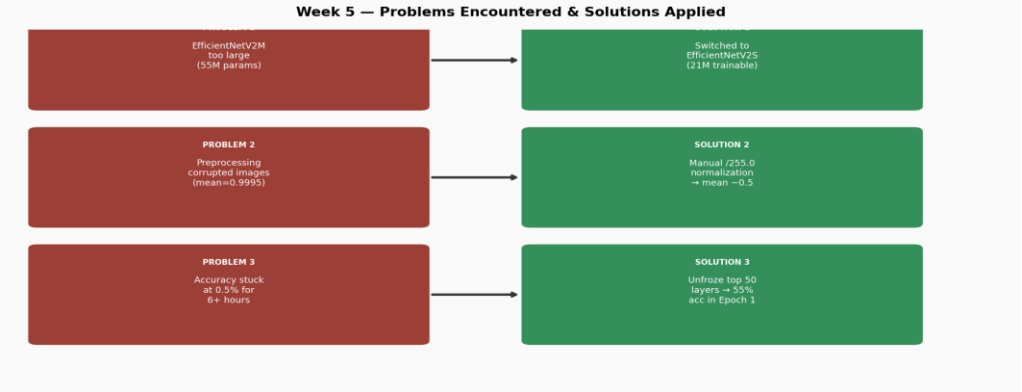

Figure 4: Three major problems encountered this week and the solutions that resolved each one

5.2 Training Results

Phase 1 training ran for 20 epochs on clean images with no degradation. Validation accuracy reached 78.03% at epoch 20. The training accuracy reached 97.15% which indicates slight overfitting addressed in the next phase through degradation fine-tuning which acts as a strong regularizer.

Metric	Value / Detail
Model	EfficientNetV2S — switched from M due to size
Total parameters	55,098,748
Trainable parameters	20,789,588 (top 50 layers + head)
Frozen parameters	34,309,160 (base backbone)

Phase 1 epochs	20 epochs on clean images
Best validation accuracy	**78.03%**
Final training accuracy	97.15%
Learning rate	1e-3 → 5e-4 (ReduceLROnPlateau)
Optimizer	Adam with categorical crossentropy loss
Model saved to Drive	**bird_model_best. keras — every best epoch**

5.3 Problems & Solutions

Problem	Solution
EfficientNetV2M caused 0.5% accuracy for 6+ hours even with clean images	Switched to EfficientNetV2S — accuracy jumped to 55% in epoch 1
Fully frozen base gave 0.68% max across 10 epochs	Unfroze top 50 layers — gave model sufficient capacity
Colab runtime reset — lost trained model	Model saved to Drive after every best epoch — resumed in minutes

5.4 Plan for Week 6

Run Phase 2 degradation fine-tuning on the saved 78.03% model, run full evaluation across all degradation severity levels, generate the comparison chart vs ChatGPT Vision and Google Lens, and build the prediction interface with quality checking.